AMARTYA RANJAN

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SUMMARY

Final-year B.Tech student in Computer Science (AI & Robotics) with production experience building data-intensive systems, automated data processing workflows, and analytics-driven ML applications. Experienced in Python, Pandas, NumPy, statistical data analysis, and large-scale data processing. Interested in Data Analytics, ETL/ELT pipelines, data platforms, distributed processing, and analytics infrastructure.

EDUCATION

Vellore Institute of Technology, Chennai 2023 – 2027
B.Tech – Computer Science (AI & Robotics) *CGPA: 8.73 / 10*
Delhi Public School, Varanasi 2023
Higher Secondary Education (XII) – CBSE *95%*

EXPERIENCE

AI Developer Intern Jan 2026 – Aug 2026
Dextora

- Built and deployed a 4-phase self-healing RAG pipeline for a client-facing production system, designing automated data ingestion, preprocessing, retrieval, evaluation, and feedback loops to improve answer relevance by 8 points over a naive RAG baseline.
- Developed data processing workflows for retrieval systems using Python, Pandas, NumPy, text preprocessing, chunking, feature extraction, and automated evaluation pipelines.
- Architected an agent orchestration system that processed structured task information and generated dispatch logic for coordinating specialised sub-agents.
- Built an end-to-end AI video generation pipeline integrating web crawlers, data extraction, FFMPEG-based media processing, and an automated scene-assembly workflow.

Research Intern 2025
CeHMA – VIT Chennai

- Designed and trained multimodal ML pipelines combining clinical records, medical imaging, genetic markers, and demographic data into a unified data processing and prediction architecture.
- Performed data preprocessing, feature engineering, heterogeneous data integration, and model development for Parkinson's disease progression prediction.
- Identified important non-motor biomarkers through data-driven analysis and contributed to an interpretable clinical decision-support framework.

PROJECTS

Self-Healing Hybrid RAG Engine *Python, Pandas, NumPy, RAG, LLM Evaluation*

- Designed a closed-loop retrieval pipeline that detects weak retrieval results, automatically triggers corrective processing, and implements feedback-driven components, improving answer relevance by 8 points over a naive RAG baseline.

High-Efficiency Deep Handwriting Transcriber *PyTorch, Bi-Axial LSTM, CTC Loss*

- Designed a 10-layer Bi-Axial LSTM pipeline for full-text handwriting transcription, achieving 11.4% Character Error Rate (CER).
- Applied spatial parallelisation to improve computational efficiency, achieving a 19× speedup over MDLSTM while maintaining accuracy.

Prodromal Parkinson's Prediction *PyTorch, Multimodal Learning, Data Processing*

- Built a multimodal data processing pipeline combining four heterogeneous data sources, feature extraction, and predictive modelling for disease progression prediction.

AI Healthcare Policy Simulator *Python, Reinforcement Learning, Census Data*

- Processed and simulated 49,736 AI household agents modelled on 2011 Indian census data to evaluate healthcare policy interventions at national scale.
- Applied data-driven reinforcement learning and decision modelling under India-specific socioeconomic constraints.

Diffusion Model from Scratch *PyTorch, Diffusion Models, VAE, U-Net*

- Implemented a diffusion-model pipeline from scratch and generated synthetic samples for downstream machine learning tasks.

TECHNICAL SKILLS

Programming: Python, C++, C, Java
Data Engineering: SQL, PostgreSQL, Apache Spark, Apache Kafka, Apache Airflow, ETL/ELT, Data Pipelines, Data Warehousing, Data Modeling, Batch Processing, Data Ingestion, Data Transformation
Cloud & Infrastructure: Docker, Linux, Git, GitHub, FFMPEG, Jupyter
Data & Analytics: Pandas, NumPy, Scikit-learn, Data Preprocessing, Feature Engineering, Statistical Data Analysis
AI / ML: PyTorch, Deep Learning, Generative AI, NLP, Multimodal Learning, Time-Series Modeling, Reinforcement Learning, Diffusion Models
AI Engineering: RAG, LLM Evaluation, LangChain, LangGraph, Agentic Systems, MCP Server Integration
Other: Foundational Blockchain Knowledge

ACHIEVEMENTS

- National-level hackathon finalist – Top 15 teams for a SEBI scam-detection model.

RESEARCH INTERESTS

Data Analytics · Data Engineering · Data Pipelines · Large-Scale Data Processing · Distributed Systems · Data Platforms · Machine Learning Systems · Time-Series Modeling · Multimodal Learning